

Daily Problem: NFA Intersection is in P

Problem Statement

Given two NFAs A_1 and A_2 , decide whether $L(A_1) \cap L(A_2) \neq \emptyset$. Prove that this problem is in P.

Solution

We solve the problem by constructing an NFA for the intersection and then checking whether its language is empty.

Step 1: Product Construction

Let $A_1 = (Q_1, \Sigma, \delta_1, s_1, F_1)$ and $A_2 = (Q_2, \Sigma, \delta_2, s_2, F_2)$. Construct an NFA A whose states are pairs (q_1, q_2) in $Q_1 \times Q_2$. The start state is (s_1, s_2) , and the accepting states are $F_1 \times F_2$.

Step 2: Correctness

The constructed automaton accepts a string w if and only if both A_1 and A_2 accept w . Thus $L(A) = L(A_1) \cap L(A_2)$.

Step 3: Emptiness Check

We perform BFS or DFS from the start state. If we reach an accepting state, the intersection is non-empty.

Step 4: Time Complexity

The product automaton has at most $|Q_1| \times |Q_2|$ states, and graph search runs in polynomial time.

Conclusion

Therefore, the NFA intersection problem is in P.